

Mechanistic Effects of the ALDH2 E487K Mutation

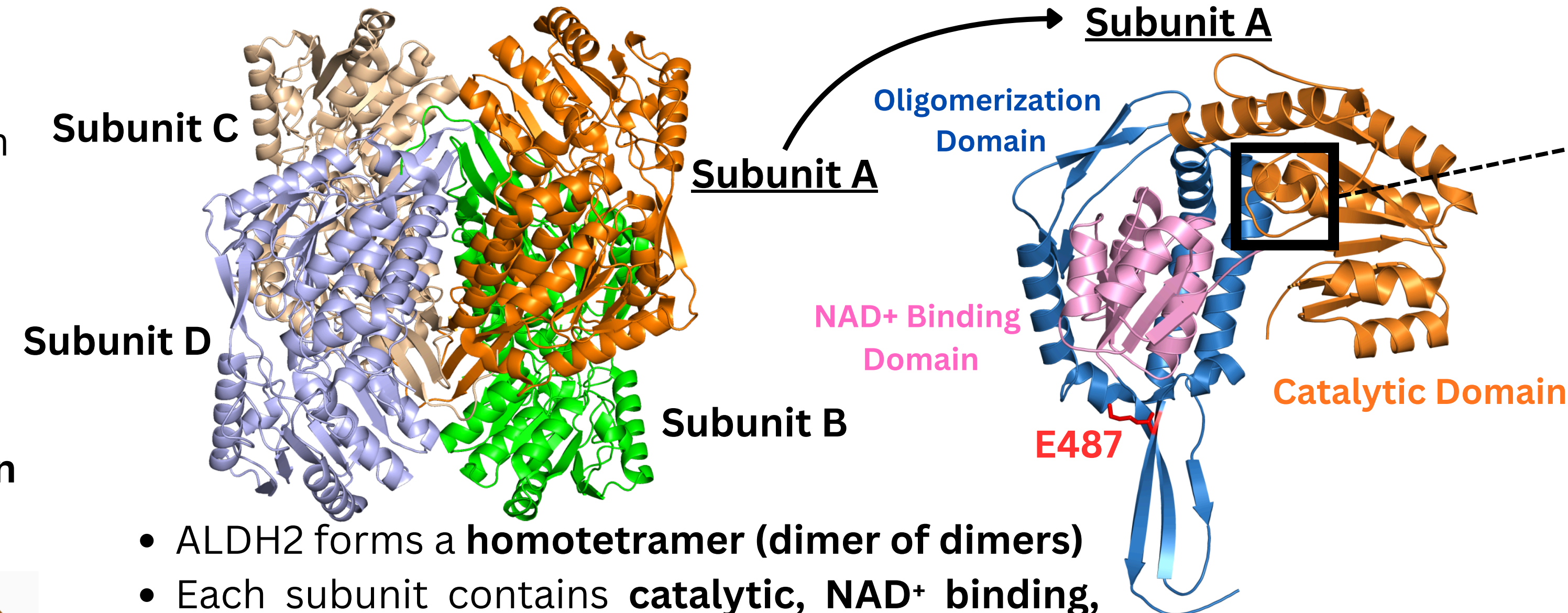
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BIOI 565 Exploring Proteins, Spring 2026

Introduction / Background

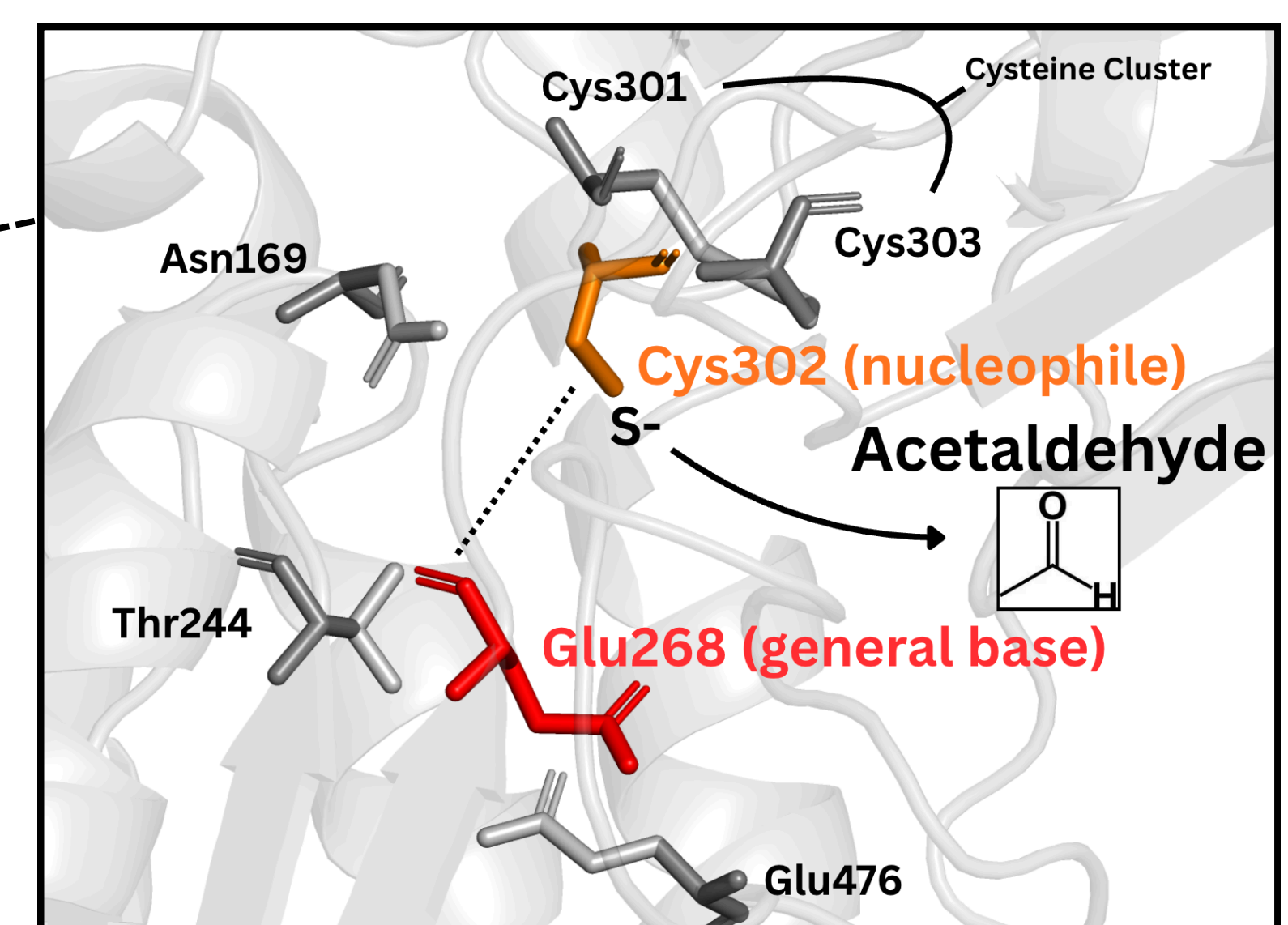
Overall Structure of ALDH2

Catalytic Mechanism & Active Site

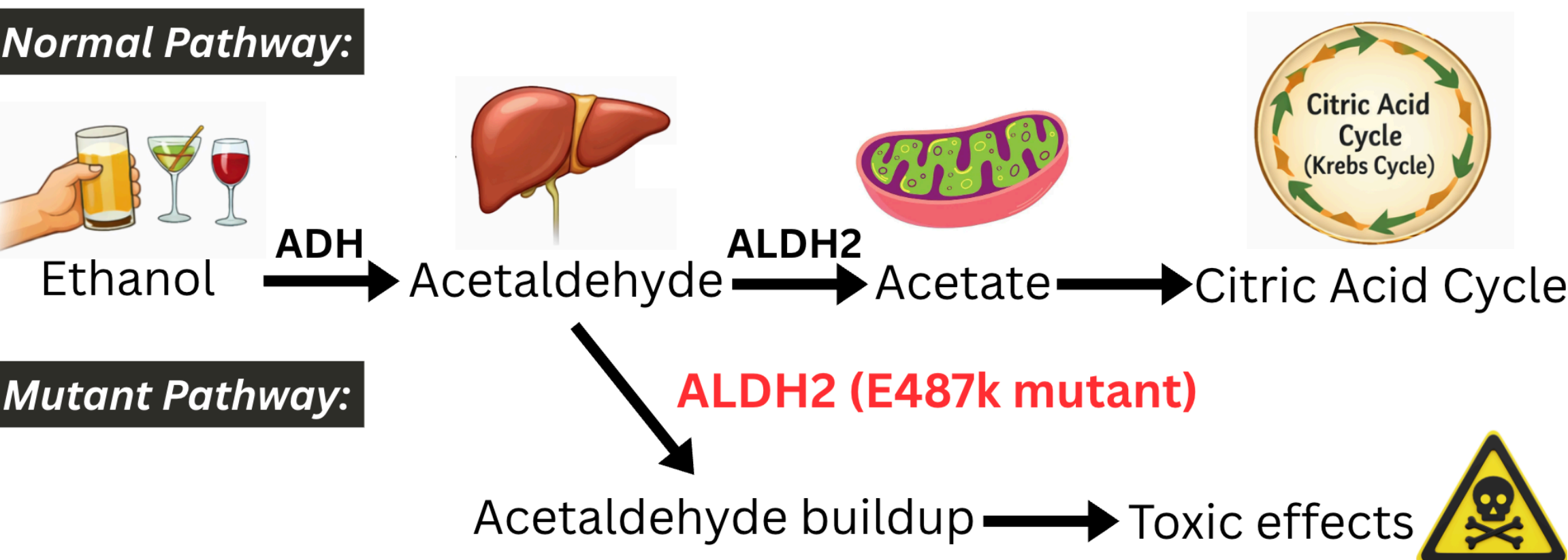
- **ALDH2** is a mitochondrial enzyme that oxidizes **acetaldehyde** → **acetate** during alcohol metabolism
- **Acetaldehyde** is highly reactive and toxic → accumulation causes flushing, nausea, and tissue damage [1, 2].
- The **E487K mutation (ALDH2*2)** significantly reduces enzymatic activity [1].
- This variant is common in East Asian populations → **impaired alcohol metabolism** [4, 5].
- This study examines how a **single amino acid substitution** **disrupts structure and function**.



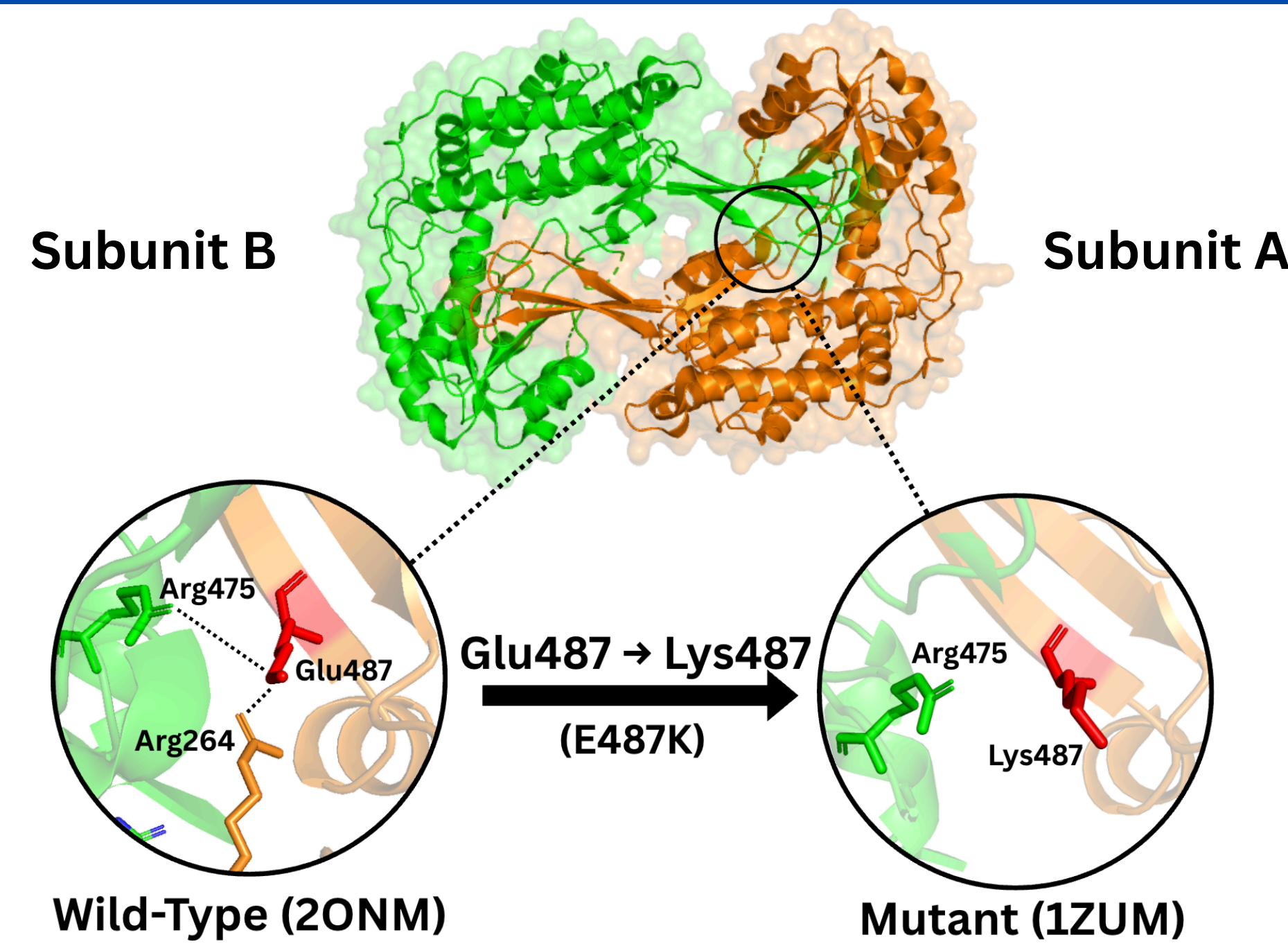
- ALDH2 forms a **homotetramer (dimer of dimers)**
- Each subunit contains **catalytic, NAD⁺ binding, and oligomerization domains**.
- **E487** is located at the **dimer interface** within the oligomerization region.
- Proper subunit interactions are required for enzyme activity.



The ALDH2 active site positions Cys302 (nucleophile) and Glu268 (general base) to drive acetaldehyde oxidation. This reaction proceeds through covalent intermediate formation, hydride transfer to NAD⁺, and hydrolysis to release acetate, as illustrated in the stepwise mechanism below:

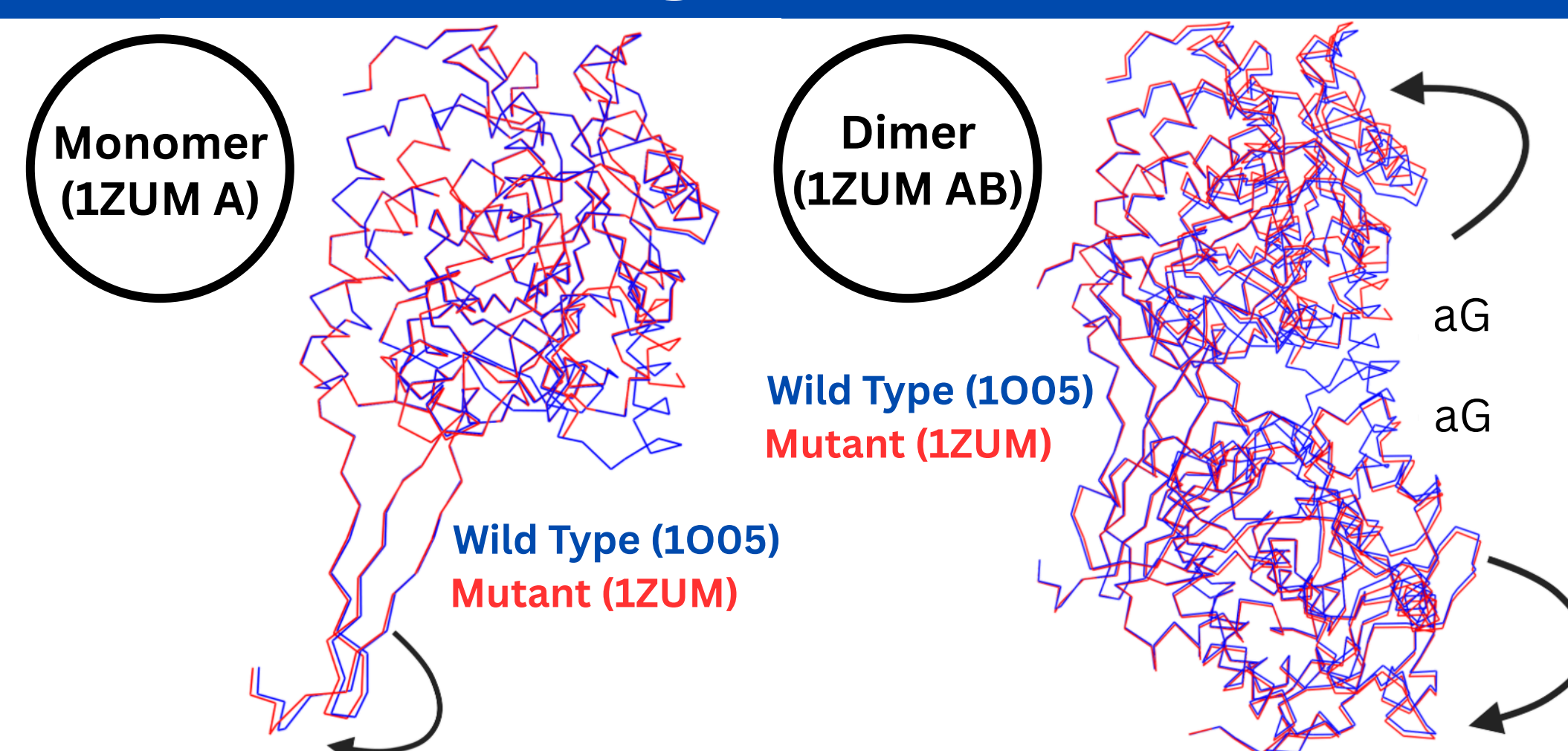


E487K Mutation



- E487K substitution** associated with the ALDH2*2 phenotype.
- Disrupts hydrogen bonding with Arg264 (same subunit) and Arg475 (adjacent subunit) [1].
 - Causes **rigid body rotations** (domain misalignment) of all domains, resulting in:
 - Decreased affinity for cofactor NAD⁺.
 - Reduced catalytic rate per subunit [1].
 - Reduction in catalytic efficiency of adjacent subunits, even WT versions [1].

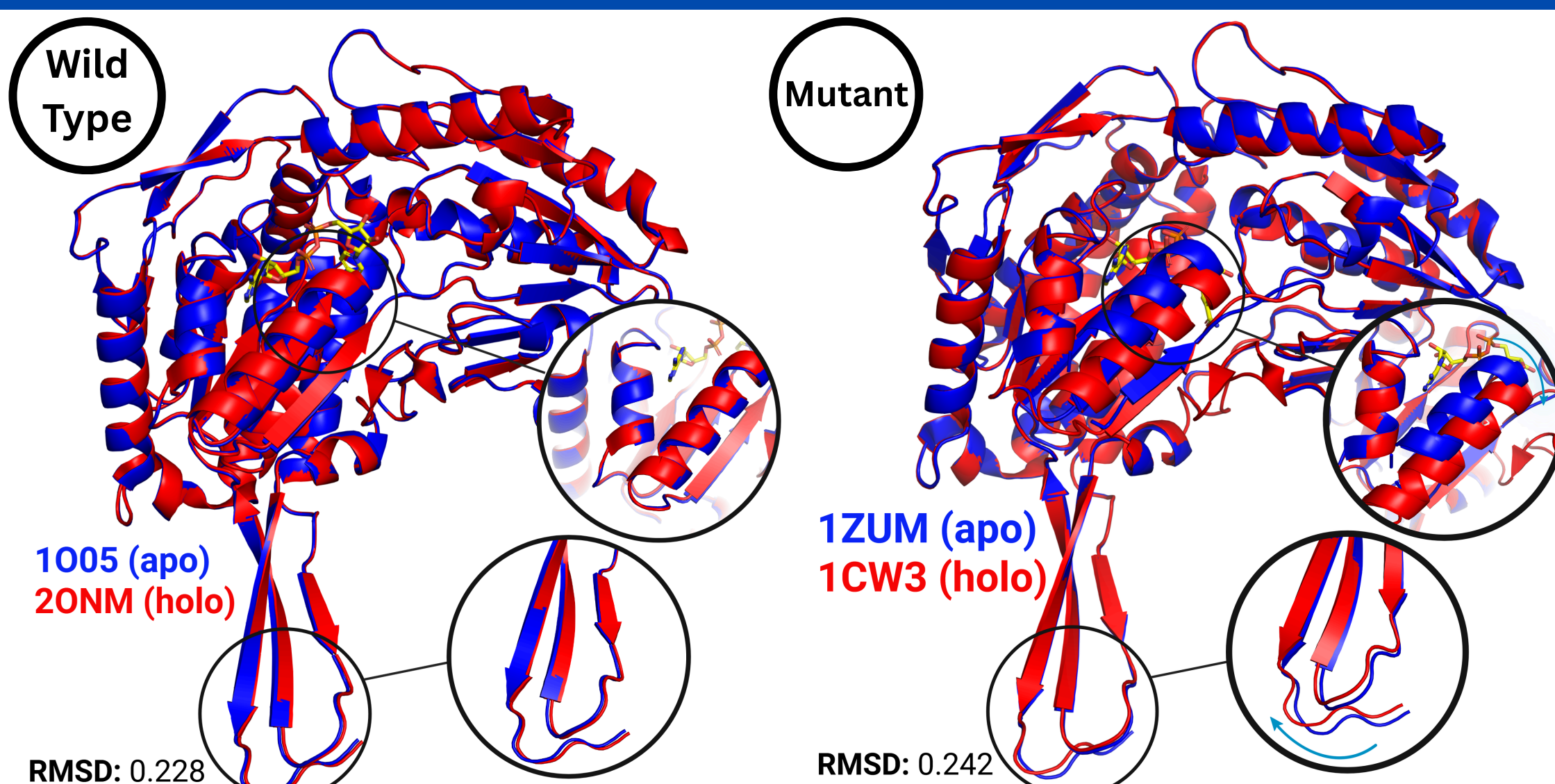
Structural Alignment (WT vs Mutant)



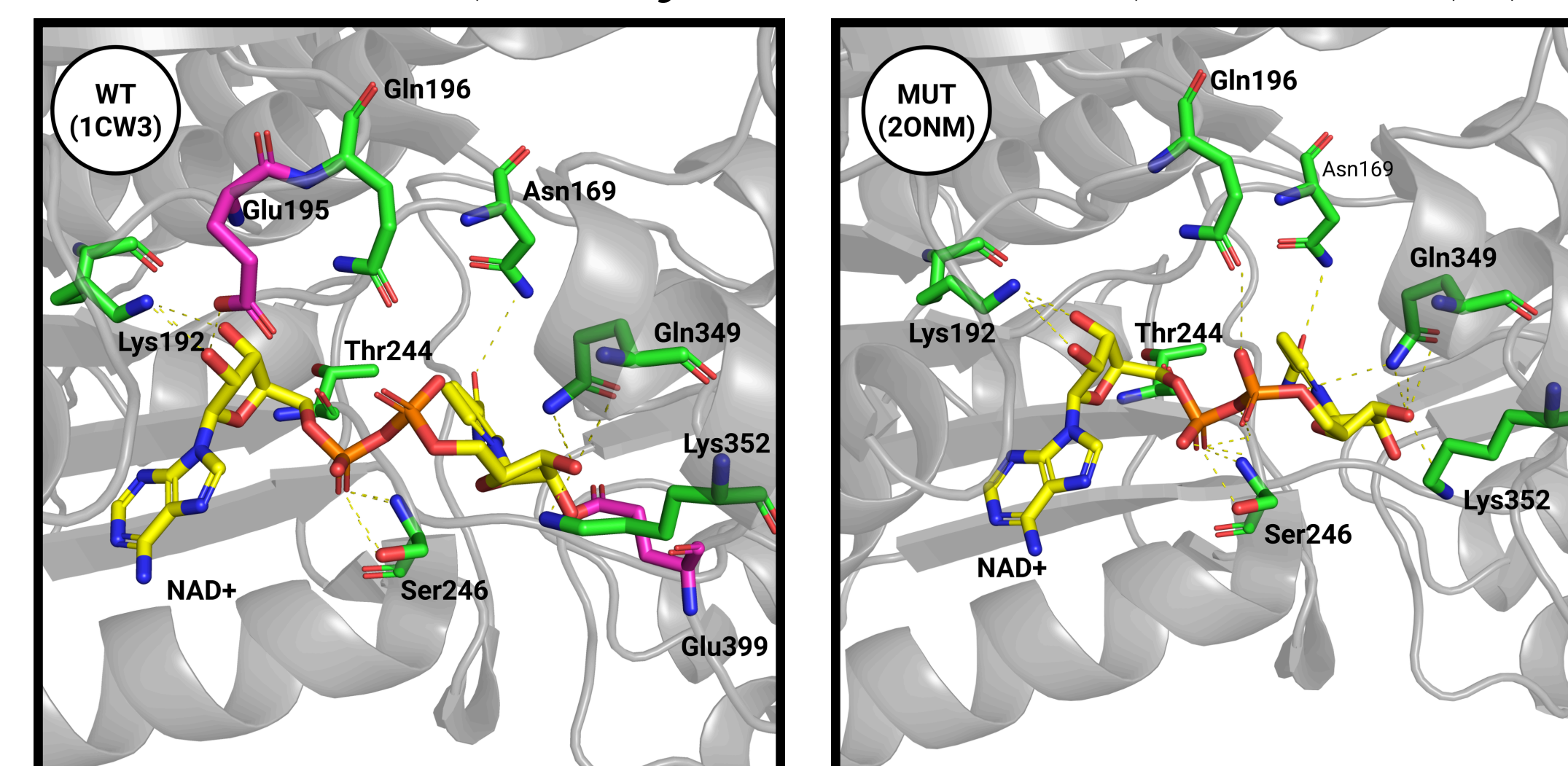
Methods

- Protein structures obtained from the Protein Data Bank
 - Mutant: 1ZUM (~2.1 Å res., apo) [1] & 1CW3 (~2.5 Å res., holo) [7].
 - Wild-type: 1005 (~2.25 Å res., apo) [2] & 20NM (~2.0 Å res., holo) [6].
- PyMOL used to visualize domain organization and residue-level differences [3].
- Structural alignment performed to identify conformational changes from the E487K substitution [3].

Impact on NAD⁺ Binding



Apo-to-holo alignment shows conformational change in wild type but substantial domain rearrangement in ALDH2*2. Pocket conserved; affinity loss is structural, not local [1, 5, 6, 7].



Functional Consequences

Normal vs. Alcohol Flush Response



Key Symptoms

- **Flushing**
- **Tachycardia**
- **Nausea/Vomiting**
- **Hypotension**

Impaired ALDH2 activity → acetaldehyde accumulation → increased alcohol sensitivity

Summary

- E487K disrupts inter-subunit interactions, altering active site geometry
- Reduces NAD⁺-dependent catalytic efficiency
- Requires greater conformational rearrangement for coenzyme binding
- Leads to acetaldehyde accumulation and clinical effects

Key insight: catalytic efficiency depends on precise subunit alignment, not just active site residues

Clinical relevance: this mutation underlies the “Asian flush” response affecting ~30–40% of East Asian populations

References

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